replace_experts.py — technical

`scripts/replace_experts.py · 170 lines · scripts/replace_experts.py`

Purpose

Initialise, replace, verify, save.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`load_and_validate_selection`

```
def load_and_validate_selection(cfg, paths) -> tuple[dict, dict]
```

lines 36-53 · raises SystemExit · settings MODEL_PATH or HF_MODEL_ID, PRUNE_EXPERTS_PER_LAYER, PRUNE_LAYERS

Load expert_selection.json and check it matches the current settings and statistics.

1. Load stats manifest and selection; the selection must match the current model source, `PRUNE_EXPERTS_PER_LAYER`, `PRUNE_LAYERS` and the statistics' calibration fingerprint; otherwise exit with the list of mismatches.

`main`

```
def main() -> None
```

lines 56-166 · raises RuntimeError, SystemExit · settings CHECKPOINT_FORMAT, PRUNE_UNOBSERVED_POLICY, REPLACEMENT_METHOD

Script entry point: runs this stage end to end.

1. Validate inputs; load statistics and the original model; check each selected layer exists, ids are valid and unique, and the layer description is unchanged since statistics collection.
2. Bias method: `initialize_bias` per selected expert (logs method, token count, norm, residuals).
3. `replace_experts` with the configured method.
4. Probe on the first 4 calibration examples with the pruned layers instrumented: gate consistency; bias -> recorded outputs equal b; drop -> outputs zero; reroute -> no token routed to a removed expert and tokens without removed experts keep identical indices and gates compared with `route(apply_mask=False)`.
5. Write `bias_initialization.json`; build the manifest; save `replaced-model/` in `CHECKPOINT_FORMAT`; log the net parameter reduction.

Errors raised

Function	Exception	Message (f-string fields in braces)
`load_and_validate_selection`	SystemExit	Stale or inconsistent artifacts - re-run score_experts.py: ...
`main`	SystemExit	Selected layer {layer_id} does not exist in the model
`main`	SystemExit	Invalid expert ids {experts} for {layer_id}
`main`	SystemExit	Layer description of {layer_id} changed since collect_stats.py
`main`	RuntimeError	{rec.layer_id}: gate consistency failed after replacement ({err})
`main`	RuntimeError	{rec.layer_id}: tokens were routed to a removed expert
`main`	RuntimeError	{rec.layer_id}: rerouting changed tokens whose experts were not removed
`main`	RuntimeError	{rec.layer_id}: dropped expert {e} produced a non-zero output